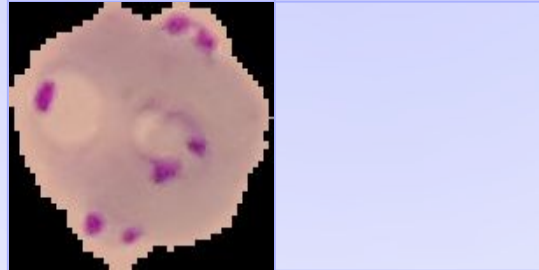
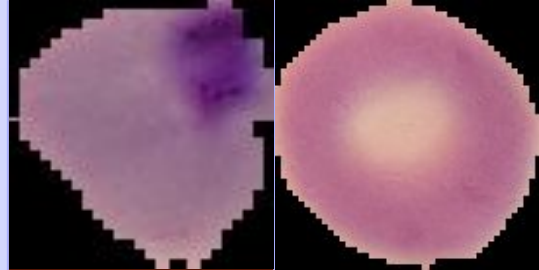


MALARIA DETECTION



AGENDA

01 Executive Summary

02 Solution Approach

03 Final Model Solution

04 Business Solution

05 Recommendations for Implementation

06 Appendix

01

EXECUTIVE SUMMARY

- Key Takeaways
- Key Next Steps

Key Takeaways

The chosen model's current performance establishes a foundation for **scalable, AI-assisted malaria detection**.

The goal is to bridge the gap between machine accuracy and **real-world medical impact**.

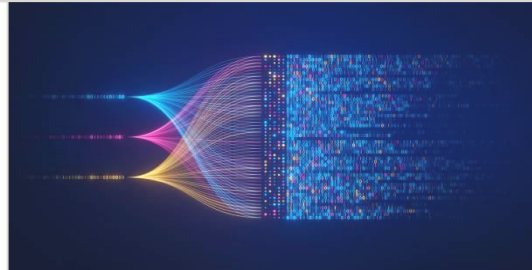
Achieving impact will require **clinical validation, ethical governance, and multi-stakeholder collaboration**.



Key Next Steps

STEP 1

Strengthen Data & Preprocessing Pipeline



STEP 2

Advance Model Robustness & Validation



STEP 3

Prepare for Scalable Deployment



SOLUTION APPROACH

- Problem Definition
- Attempted CNN Models

Evaluation Metrics

- Accuracy
- Precision, Recall, and F1-score
- Confusion matrix (FP & FN)
- Train-Validation gap

Problem Definition

01

Diagnostic Challenge

Manual microscopy is **slow**, **error-prone**, and **dependent** on expert availability.

02

Technical Objective

Automate binary classification ('Parasitized' vs 'Uninfected') to deliver **consistent**, **scalable**, and **expert-level** screening.

03

Business & Clinical Need

Develop an AI system that improves **diagnostic accuracy**, **efficiency**, and **accessibility** in real-world healthcare settings.

Attempted CNN Models

Base Model

Simple CNN trained from scratch.

- Strong accuracy and sensitivity
- Efficient and stable baseline

Model 1: Extra Layers

Added depth to increase complexity.

- Most sensitive (low FN)
- Longer training yet no significant accuracy gain

Model 2: BatchNorm+LeakyReLU

Added normalization and smoother activation for stability.

- Most conservative (low FP)
- Improved regularization but higher false negatives

Model 3: Data Augmentation

Introduced rotation, zoom, and flip transformations.

- Best overall generalization and robustness
- High accuracy

VGG16: Transfer Learning

Used pre-trained ImageNet features with custom classifier layers.

- Fast convergence, strong stability, but mild overfitting
- Good trade-off between speed and accuracy

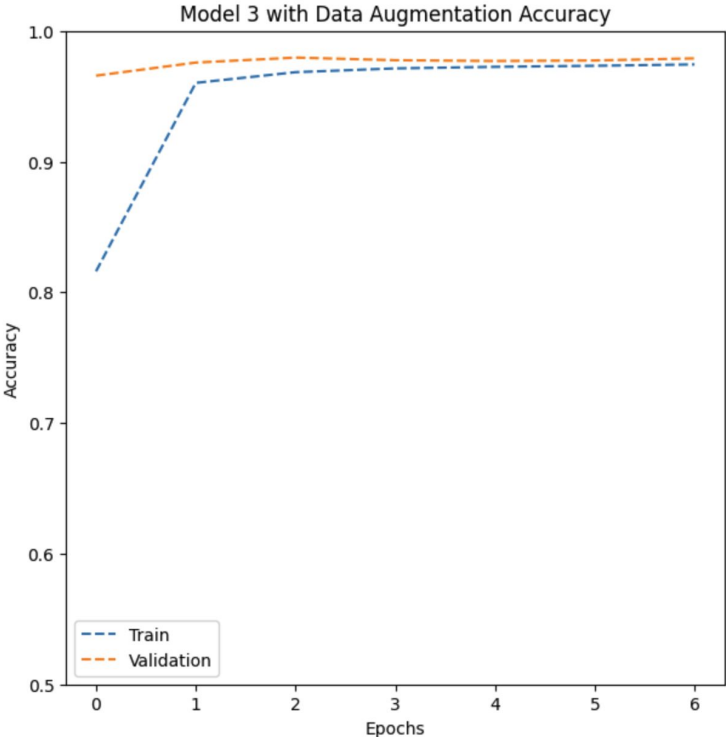
03

FINAL MODEL SOLUTION:

Model 3: Data Augmentation

- High Accuracy
- Best Generalization (Train $\approx$ Val)
- Practical Reliability

Performance Metrics



98.42%

Model Accuracy

0.98

F1 Score

0.98

Precision

0.98

Recall

0.50%

False Positives

1.08%

False Negatives

Architecture Pipeline

01 Input

64×64×3 RGB microscope images (rescaled to [0, 1]).

02 Augmentation

On-the-fly transformations: horizontal flip, $\pm 30^\circ$ rotation, 0.2 zoom.

- ✓ Increases image variability and reduce overfitting.

03 Feature Extraction

3 Convolution blocks (each Conv2D(32, 3×3, ReLU, same) → MaxPool(2×2) → Dropout(0.2).

- ✓ Captures cell shape and texture features.

04 Classifier Head

Flatten → Dense(512, ReLU) → Dropout(0.4) → Dense(2, Softmax)

- ✓ Outputs probabilities for 'Uninfected' and 'Parasitized' cells.

05 Training Strategy

Adam optimizer (LR=0.001), Binary cross-entropy loss, Early stopping + Checkpointing.

- ✓ Achieves high test accuracy with strong generalization.

04

BUSINESS SOLUTION

- Proposed Business Solution
- Executing Business Solution

Proposed Business Solution

AI-Assisted Malaria Screening Platform

Leverage the Model 3 pipeline as a cloud/edge inference service for automatic classification of 'Parasitized' vs 'Uninfected' cells, which:

- Accelerates early diagnosis
- Enables faster triage
- Supports medical staff in regions with limited lab capacity

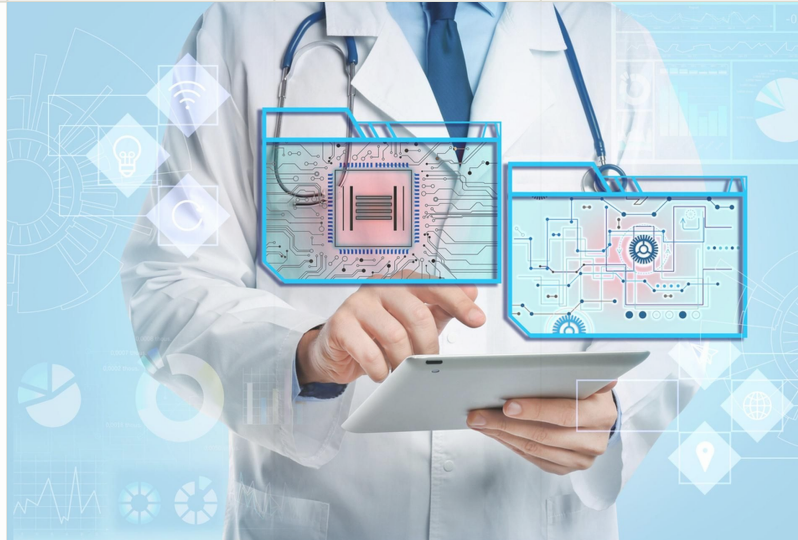


Executing Business Solution

STEP 1

Model Validation & Clinical Benchmarking

- Partner with hospitals/labs to test the model on real-world datasets
- Evaluate sensitivity against expert pathologists
- Adjust thresholds to reduce FN (critical in clinical use)



Executing Business Solution

STEP 2

Product Integration & Interface Design

- Build a web-based dashboard/mobile app that:
 - Uploads microscope images directly/through connected slide scanners
 - Returns infection probability with visualization
 - Allows overriding and annotating
- Integrate the trained model as an API service



Executing Business Solution

STEP 3

Deployment Architecture



- Two-tier deployment:
 - Cloud inference for centralized analysis
 - Edge inference for low-connectivity clinics
- Use automated retraining pipeline to continuously update the model with new annotated data

Executing Business Solution

STEP 4

Regulatory Compliance & Ethical Framework

- Ensure compliance with WHO and FDA digital-diagnostic validation requirements
- Establish data governance for patient privacy
- Include human-in-the-loop review before any automated clinical decision



Executing Business Solution

STEP 5

Pilot Program & Scale-Up

- Launch pilot regional hospitals to monitor performance in routine workflow
- Collect user feedback from labs and clinics
- Iterate model tuning and interface usability, then scale to broader geographies



05

RECOMMENDATIONS FOR IMPLEMENTATION

- Key Actionables for Stakeholders
- Risks & Challenges
- Comparison of Malaria Detection Methods

Key Actionables for Stakeholders

	Healthcare Institutions & Labs	Government & Health Ministries	Medical Professionals & Researchers	Technology Teams / Developers
Actionable Items	■ Integrate AI-assisted malaria screening in routine workflows ■ Pilot in diagnostic labs with existing microscopy setups	■ Define regulatory frameworks for AI-aided diagnostics ■ Establish quality assurance protocols	■ Validate and retrain models with locally sourced, diverse data ■ Provide domain feedback	■ Extend pipeline by building a pre-cleaning model ■ Expand dataset diversity
Benefits	■ Reduced diagnostic time and workload	■ Strengthened public health infrastructure	■ Improved accuracy across demographic and geographic variations	■ More robust deployment and trustworthiness
Costs	■ Moderate setup cost for integration and image preprocessing	■ Upfront investment in compliance and digital capacity building	■ Training and annotation costs	■ Higher computational complexity during development

Risks

External dependencies and issues that can derail deployment or cause harm if not properly managed.

01

Data Labeling Bias

Mislabelled or inconsistent data may propagate diagnostic errors and reduce model reliability.

02

Privacy & Ethical Governance

Weak governance or data privacy breaches can delay adoption and raise legal concerns.

03

Regulatory Validation & Compliance

Delays or failures in meeting WHO/FDA standards can block clinical deployment.

Challenges

Technical or operational barriers that require optimization, adaptation, or collaboration to overcome.

01

Performance Saturation

Pushing accuracy beyond 98% faces diminishing returns.

02

Unrealistic Dataset

Actual raw dataset contains staining inconsistencies, debris, lighting variations, and partially focused cells.

03

Operational Integration & Infrastructure

Requires consistent imaging quality and adequate digital infrastructure across sites.

04

Interpretability & Clinical Trust

Limited transparency may hinder clinician trust without explainability tools.

Comparison of Malaria Detection Methods

	DL (Microscopy)	RDT	PCR
Strengths	■ High accuracy ■ Scalable, automated ■ Cost-efficient post-deployment	■ Fast ■ Easy to use ■ Cheap	■ Extremely accurate ■ Species-specific
Weaknesses	■ Needs digital imaging setup ■ Sensitive to smear quality ■ Regulatory hurdles	■ Lower accuracy (~84%) ■ Ambiguity information	■ Slow ■ Lab-dependent ■ Expensive
Best use case	■ Hospital/lab AI integration ■ Telepathology	■ Field screening ■ Community outreach	■ Confirmatory testing ■ Research ■ Surveillance

THANK YOU

Any questions or comments are welcome.

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Model Comparison

Model Accuracy Comparison Summary (Descending Order)

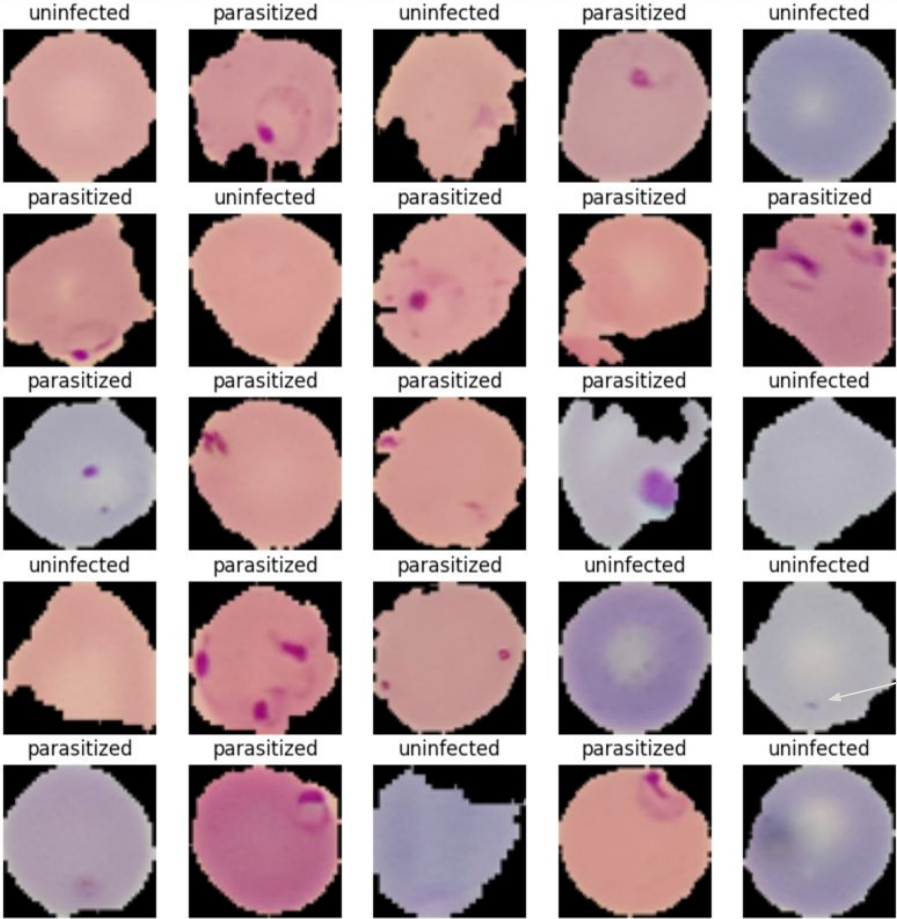
	accuracy	FP	FN
Model 3	0.9842	13	28
VGG16	0.9835	35	8
Base	0.9827	34	11
Model 2	0.9785	7	49
Model 1	0.9608	101	1

Best Performers by Category

Metric / Aspect	Best Model	Reason
Highest Accuracy	Model 3	Highest test accuracy among all models.
Lowest False Positives (FP)	Model 2	Most conservative classifier; Batch Normalization minimized false positives.
Lowest False Negatives (FN)	Model 1	Most sensitive model for detecting parasitized cells — virtually no misses.
Best Generalization (Train ≈ Val)	Model 3	Exhibited the smallest train–validation gap and smooth convergence.
Fastest Convergence	VGG16	Reached stable accuracy (~0.98) within 2–3 epochs, indicating rapid representational alignment.
Best Runtime Efficiency	Base Model	Lightest architecture with fast epochs and minimal overhead.
Most Balanced Trade-off	Model 3	Optimal compromise among accuracy, sensitivity, precision, and generalization stability.

■ Notes taken to easier compare the 5 models.

Risk #1



Data Labeling Bias:
How to deal with
labeling edge cases

Building Code for Model 3

Use image data generator

```
from sklearn.model_selection import train_test_split

X_train, X_val, y_train, y_val = train_test_split(train_images, train_labels, test_size = 0.2, random_state = 42)

from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Using ImageDataGenerator to generate images
train_datagen = ImageDataGenerator(horizontal_flip = True, zoom_range = 0.2, rotation_range = 30)
val_datagen = ImageDataGenerator()

# Flowing training images using train_datagen generator
train_generator = train_datagen.flow(x = X_train, y = y_train, batch_size = 64, seed = 42, shuffle = True)

# Flowing validation images using val_datagen generator
val_generator = val_datagen.flow(x = X_val, y = y_val, batch_size = 64, seed = 42, shuffle = True)
```

Building the model

```
from keras.models import Sequential
from keras.layers import (Input, Conv2D, BatchNormalization, LeakyReLU, MaxPooling2D, Dropout, Flatten, Dense)

# Creating sequential model
model = Sequential(name="model_3")

# 1st Convolutional layer
model.add(Conv2D(filters = 32, kernel_size = 3, padding = "same", activation = "relu", input_shape=(64, 64, 3)))
model.add(MaxPooling2D(pool_size = 2))
model.add(Dropout(0.2))

# 2nd Convolutional layer
model.add(Conv2D(filters = 32, kernel_size = 3, padding = "same", activation = "relu"))
model.add(MaxPooling2D(pool_size = 2))
model.add(Dropout(0.2))

# 3rd Convolutional layer
model.add(Conv2D(filters = 32, kernel_size = 3, padding = "same", activation = "relu"))
model.add(MaxPooling2D(pool_size = 2))
model.add(Dropout(0.2))

# Flatten and Dense layers
model.add(Flatten())
model.add(Dense(512, activation = "relu"))
model.add(Dropout(0.4))

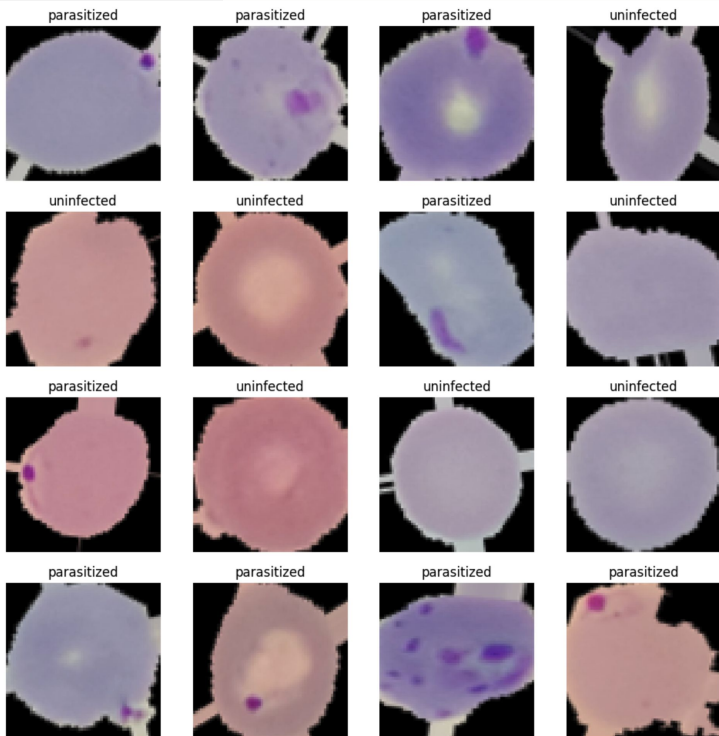
# Output layer
model.add(Dense(2, activation = "softmax")) # 2 represents output layer neurons
```

- Using image data generator
- Building the model (same as Base model)

Visualizing Augmented images

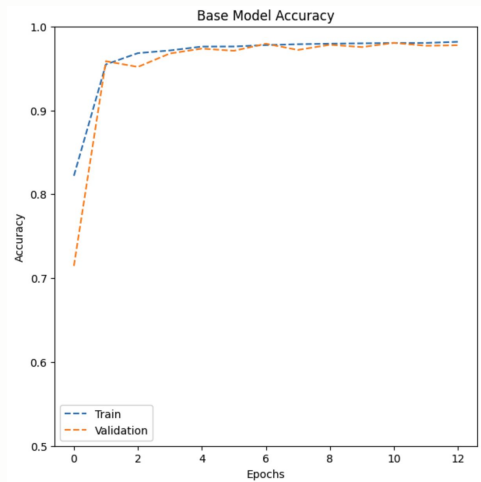
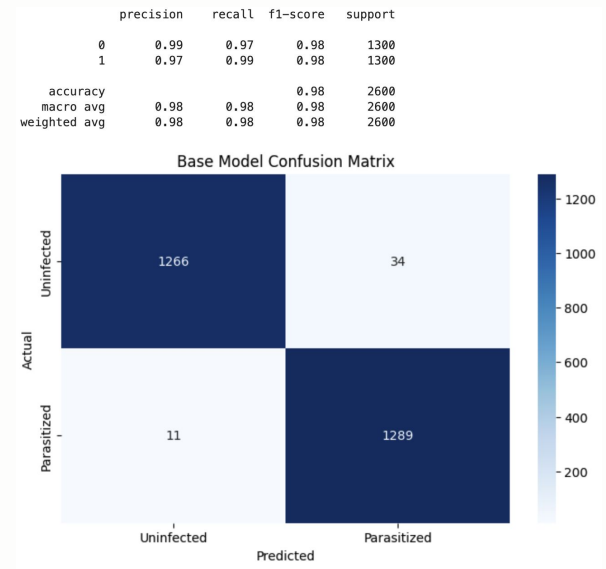
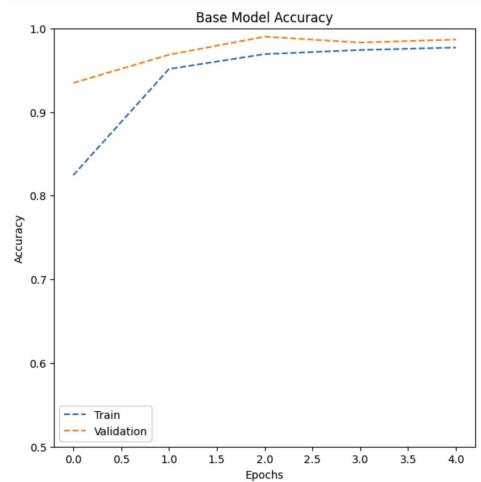
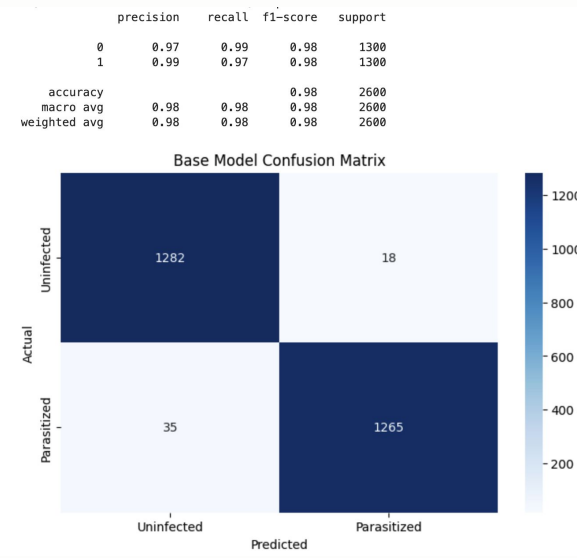
```
# Creating an iterable for images and labels from the training data
images, labels = next(train_generator)
```

```
# Plotting 16 images from the training data
fig, axes = plt.subplots(4, 4, figsize = (16, 8))
fig.set_size_inches(12, 12)
for (image, label, ax) in zip(images, labels, axes.flatten()):
    ax.imshow(image)
    if label[1] == 1:
        ax.set_title('parasitized')
    else:
        ax.set_title('uninfected')
    ax.axis('off')
```



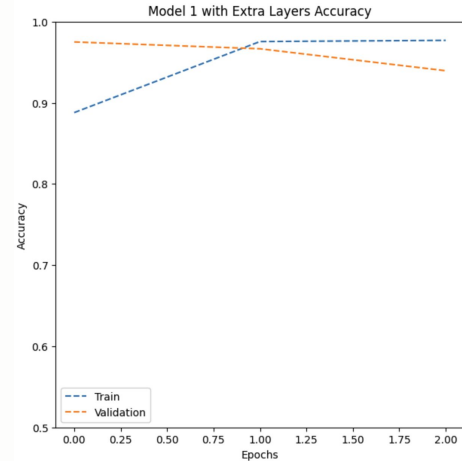
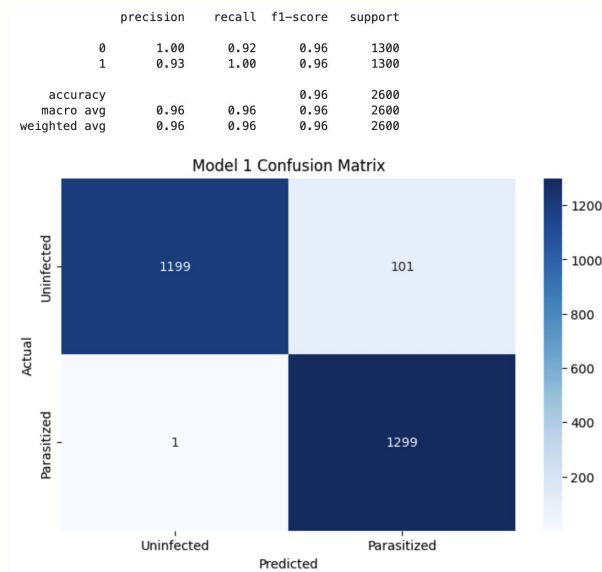
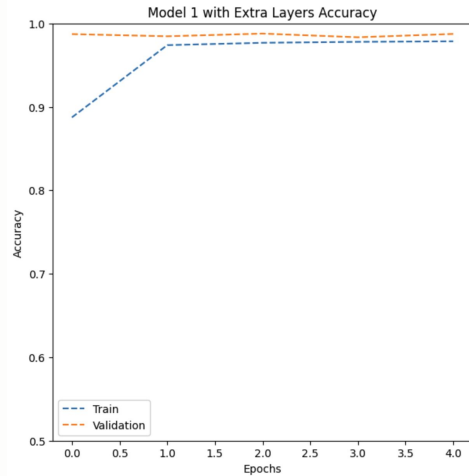
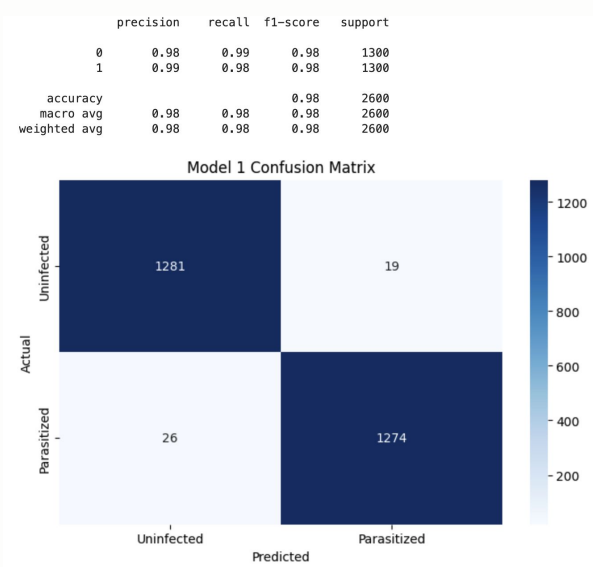
■ Visualizing Augmented Images

Base Model



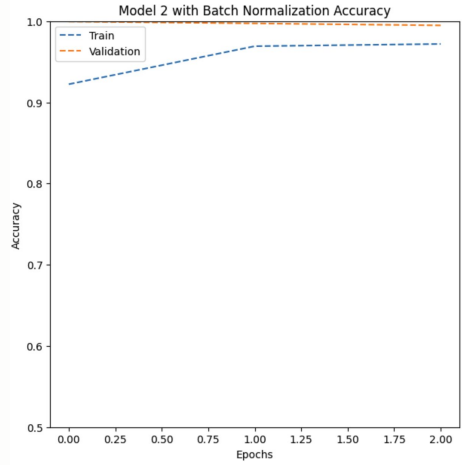
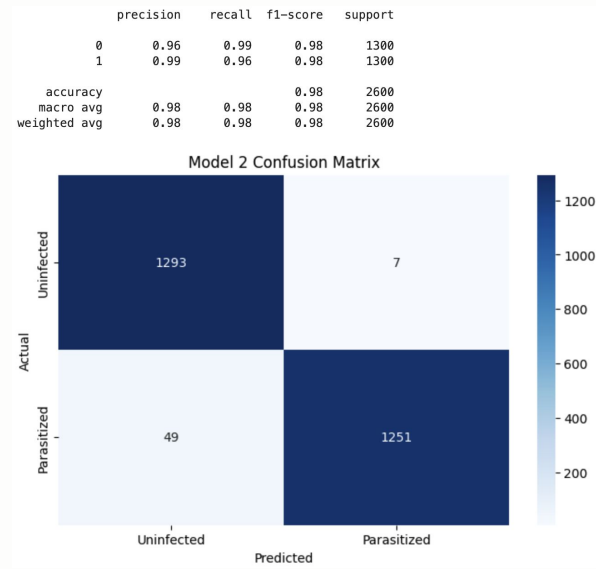
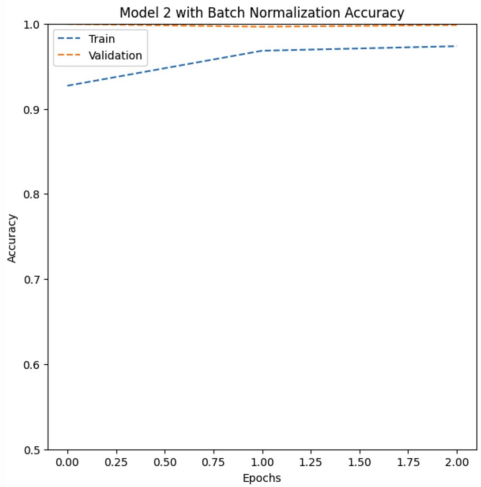
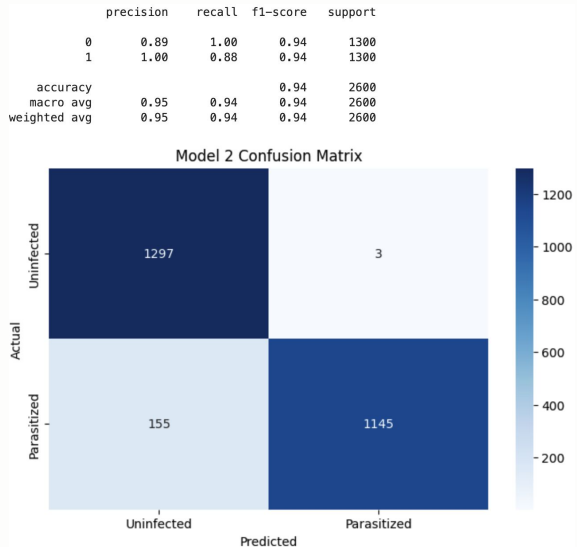
- accuracy: 97.96%–98.27%
- FP: 18~34
- FN: 11~35

Model 1: Extra Layers



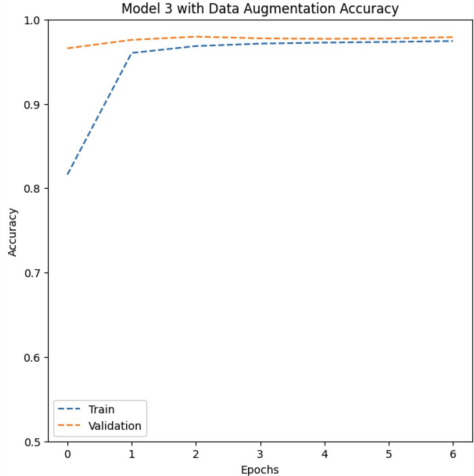
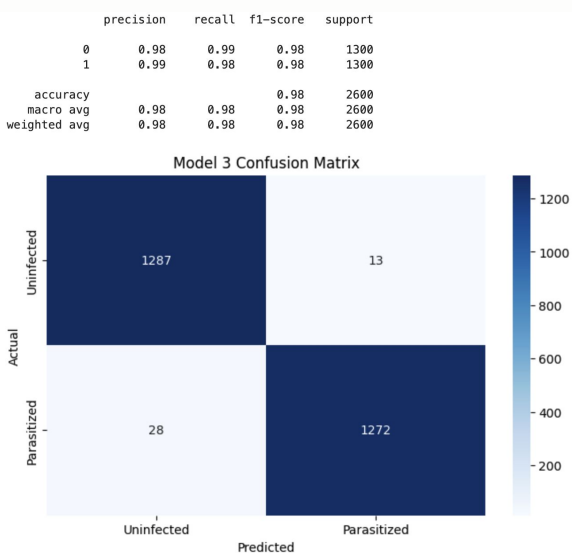
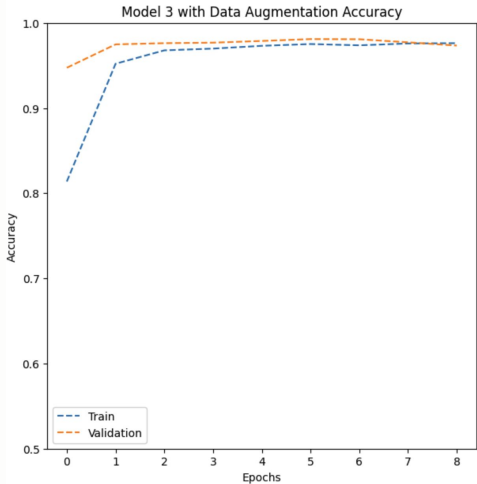
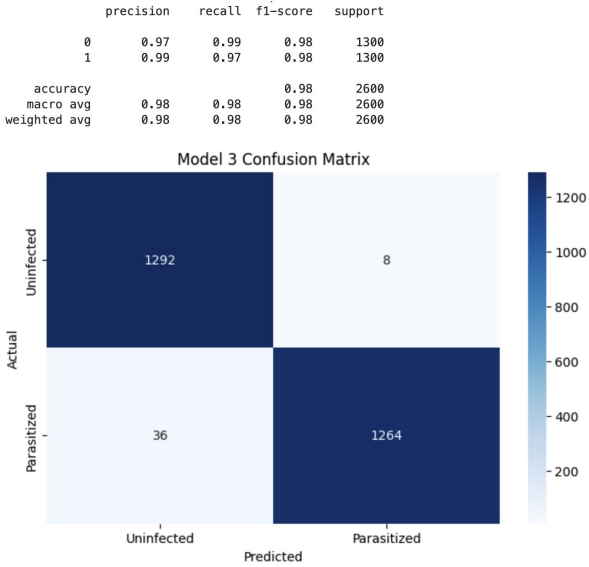
- accuracy:
96.08%-98.27%
- FP: 19-101
- FN: 1-26

Model 2: BatchNorm + LeakyReLU



- accuracy: 93.92%-97.85%
- FP: 3-7
- FN: 49-155

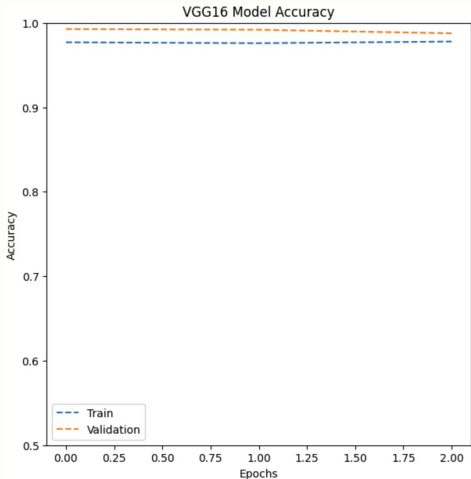
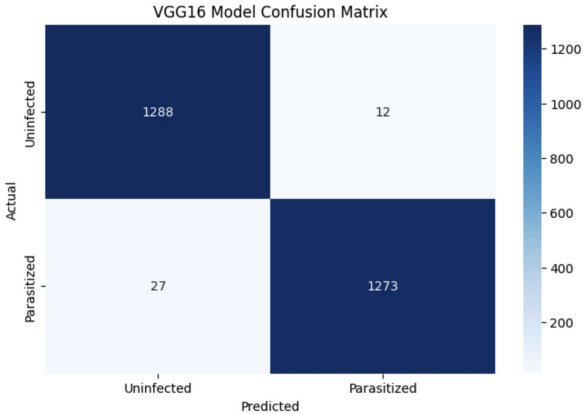
Model 3: Data Augmentation



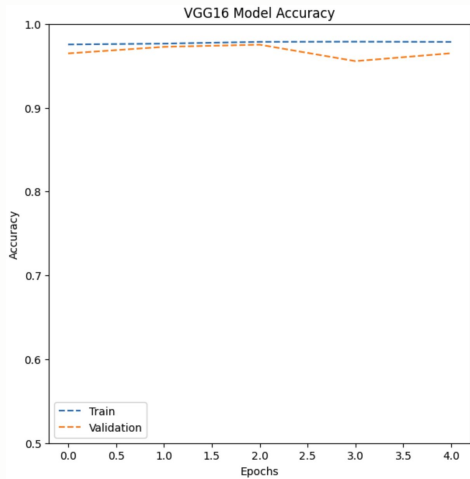
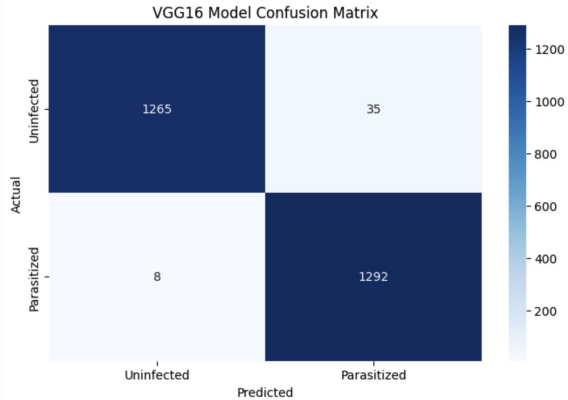
- accuracy: 98.31%-98.42%
- FP: 8-13
- FN: 28-36

VGG16: Transfer Learning

	precision	recall	f1-score	support
0	0.98	0.99	0.99	1300
1	0.99	0.98	0.98	1300
accuracy			0.98	2600
macro avg	0.99	0.98	0.98	2600
weighted avg	0.99	0.98	0.98	2600



	precision	recall	f1-score	support
0	0.99	0.97	0.98	1300
1	0.97	0.99	0.98	1300
accuracy			0.98	2600
macro avg	0.98	0.98	0.98	2600
weighted avg	0.98	0.98	0.98	2600



- accuracy: 98.35%-98.50%
- FP: 12-35
- FN: 8-27